
CAN WE TRUST LLM SELF-EXPLANATIONS?

AN EMPIRICAL EVALUATION OF EXPLANATION-SUPPORT FAITHFULNESS ACROSS REASONING TASKS

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ABSTRACT

Large Language Models (LLMs) can generate explanations alongside their answers, but it is not always clear whether these explanations provide adequate support for the answers they produce. This paper investigates the faithfulness of self-generated explanations from three open-weight models: Qwen 2.5 3B, Llama 3.2 3B, and DeepSeek-R1 Distill Qwen 1.5B.

The models were evaluated on three reasoning benchmarks: GSM8K, CommonsenseQA, and StrategyQA. Each model answered 100 fixed questions from each dataset, resulting in 900 responses. Answer accuracy, output format compliance, and generation time were measured automatically. In addition, a balanced sample of 90 responses was manually evaluated for explanation-support faithfulness, logical consistency, completeness, and common failure types.

The results show clear differences across models and tasks. DeepSeek achieved the highest accuracy on GSM8K (80%), while Qwen performed best on CommonsenseQA (76%) and StrategyQA (60%). Llama followed the required output format in all cases, whereas DeepSeek achieved 85.67% strict-format compliance. In the manually evaluated sample, explanations accompanying correct answers received a mean faithfulness score of 2.64, compared with 0.89 for incorrect answers ($U = 1827.0$, $p < 0.001$, Cliff's $\delta = 0.804$).

Despite these differences, all three models produced explanations containing unsupported claims, reasoning errors, or inconsistencies between the explanation and the final answer. These findings suggest that self-generated explanations can help assess a model's observable reasoning behavior, but their plausibility alone is not sufficient evidence of faithfulness. Importantly, this study evaluates how well explanations support their answers, rather than whether they reflect the models' internal reasoning processes.

Keywords Explainable Artificial Intelligence · Large Language Models · Self-Explanations · Explanation Faithfulness

GitHub Repository: <https://github.com/Yash-Gavade/FaithfulLLM>

1 Introduction

Large Language Models (LLMs) are widely used for tasks that involve reasoning and decision-making. Many of these models can also provide natural-language explanations together with their answers. This allows users to inspect how an answer is justified instead of seeing only the final prediction. However, an explanation that seems convincing is not necessarily correct or reliable. It may contain unsupported claims, reasoning errors, or statements that contradict the predicted answer [Miller, 2019; Schneider, 2024].

This creates an important distinction between explanation plausibility and faithfulness. A plausible explanation may seem reasonable to a reader without accurately representing the factors that influenced the model’s prediction. Previous research has shown that language models can generate explanations that do not consistently reflect the information influencing their answers [Lanham et al., 2023; Madsen et al., 2024; Turpin et al., 2023]. Therefore, an explanation should not be considered faithful only because it is fluent or convincing.

Evaluating faithfulness is more difficult than evaluating whether an explanation appears plausible. Some studies investigate this by modifying parts of the input or intervening on generated reasoning and then observing whether the model’s behavior changes. Other studies examine whether the evaluation methods themselves provide reliable evidence of faithfulness [Parcalabescu and Frank, 2024; Zaman and Srivastava, 2025]. Since these approaches evaluate faithfulness in different ways, it is important to define clearly what is being measured in this paper.

In this study, I focus on *explanation-support faithfulness*. I use this term to describe how well the written explanation supports the final answer reported by the model. A good explanation should provide relevant information, follow consistent reasoning, and avoid unsupported conclusions. However, this definition is limited to the explanation that can be observed in the generated output. It does not show whether the explanation represents the model’s actual internal reasoning process.

Previous studies have reported several problems with LLM-generated explanations, including unfaithful reasoning and disagreement between an explanation and the factors influencing a model’s prediction [Madsen et al., 2024; Matton et al., 2025; Paul et al., 2024]. In this paper, I compare three open-weight language models under the same experimental setup. The aim is to examine how their answers and explanations differ across several reasoning tasks when the same questions, prompt structure, and evaluation procedure are used.

The three evaluated models are Qwen 2.5 3B, Llama 3.2 3B, and DeepSeek-R1 Distill Qwen 1.5B [DeepSeek-AI, 2025; Grattafiori et al., 2024; Yang et al., 2024]. They are evaluated on GSM8K, CommonsenseQA, and StrategyQA. These datasets represent three different types of reasoning: mathematical reasoning, commonsense question answering, and multi-hop yes/no reasoning [Cobbe et al., 2021; Geva et al., 2021; Talmor et al., 2019].

For each dataset, I selected 100 questions and used the same questions for all three models. Each model was asked to provide a final answer together with a short explanation. This resulted in 900 responses in total. Answer accuracy, output-format compliance, and generation time were evaluated automatically. I also manually reviewed a balanced sample of 90 responses to examine explanation-support faithfulness, logical consistency, completeness, and common explanation failures.

The study addresses the following research questions:

1. **RQ1:** How does answer accuracy vary across the three models and reasoning tasks?
2. **RQ2:** How consistently do the models follow the required answer–explanation format?
3. **RQ3:** How does explanation-support faithfulness vary across models, datasets, and correct versus incorrect answers?
4. **RQ4:** What types of explanation errors appear in the manually reviewed responses?

The evaluation is based on three hypotheses:

- **H1:** Model accuracy will vary across the three reasoning tasks.
- **H2:** Explanations accompanying correct answers will receive higher explanation-support faithfulness scores than explanations accompanying incorrect answers.
- **H3:** The reasoning-oriented model may achieve higher accuracy on the mathematical reasoning task while requiring more generation time and showing lower format compliance.

The main contribution of this paper is an empirical comparison of three open-weight models across three different reasoning tasks. The evaluation combines automatic measurements of answer accuracy, format compliance, and generation time with a manual analysis of explanation quality. The automatic evaluation covers 900 responses, while 90 responses are examined in more detail. This makes it possible to study not only whether the models produce correct answers, but also whether their explanations provide reasonable support for those answers and what kinds of problems appear in the generated explanations.

2 Related Work

2.1 XAI and LLM Self-Explanations

Explainable Artificial Intelligence (XAI) aims to make the behavior of AI systems easier for people to understand. Traditional approaches such as LIME and SHAP estimate how different input features contribute to a model’s prediction [Lundberg and Lee, 2017; Ribeiro et al., 2016]. However, explainability is not only about identifying important features. An explanation should also help a user understand why a particular prediction was made [Lipton, 2018; Miller, 2019].

With LLMs, explanations can be generated directly in natural language. One well-known example is Chain-of-Thought prompting, where the model is encouraged to produce intermediate reasoning steps before reaching an answer. Self-consistency extends this idea by considering answers from multiple reasoning paths [Wang et al., 2023; Wei et al., 2022]. Other approaches, including STaR and Self-Refine, use generated reasoning or feedback to improve later responses [Madaan et al., 2023; Zelikman et al., 2022].

These methods show that generated reasoning can be useful for improving model performance. Still, an improvement in accuracy does not necessarily mean that the generated explanation faithfully describes the reasoning behind the answer. This becomes important when explanations are used not only to improve performance, but also to understand or evaluate a model’s output.

2.2 Evaluating Explanation Faithfulness

One of the main problems in evaluating LLM explanations is that a plausible explanation is not always a faithful one. A response may be fluent, detailed, and convincing to a reader while still failing to represent the factors that influenced the model’s prediction.

Previous studies have shown this problem in different ways. Turpin et al. (2023), for example, found that biases introduced into prompts can influence model answers without being mentioned in the generated reasoning. Lanham et al. (2023) use interventions on generated reasoning and observe whether the model’s behavior changes. These studies show why the quality of the written explanation alone cannot establish whether it is faithful.

Other work has examined self-generated explanations and their relationship with model predictions [Madsen et al., 2024; Matton et al., 2025; Paul et al., 2024]. A common issue in this work is that an explanation can look informative and well reasoned without necessarily providing reliable evidence about why the model produced a particular answer.

Related research has also studied hallucination and uncertainty in generated text. SelfCheckGPT compares multiple generated responses to identify possible hallucinations, while semantic uncertainty methods consider differences in the meaning of generated outputs [Farquhar et al., 2024; Manakul et al., 2023]. These methods are useful for checking the reliability of generated content, but they focus on a different problem from the one examined in this paper. They do not directly measure whether a written explanation provides adequate support for the final answer reported by the model.

Another direction is mechanistic interpretability, which studies internal model components and learned features [Bricken et al., 2023; Elhage et al., 2021]. This differs from the evaluation used in this paper. Here, I only examine the explanation that appears in the generated response. I do not inspect the internal computations that produced the answer or explanation.

2.3 Research Gap

Previous research provides strong evidence that LLM-generated explanations can be useful but are not always reliable. At the same time, the existing studies differ in the models they evaluate, the reasoning tasks they use, their prompting methods, and even how they define faithfulness. Because of these differences, their results are not always easy to compare directly.

In this paper, I use the same experimental setup for three open-weight models and evaluate them across mathematical, commonsense, and multi-hop reasoning tasks. The models receive the same selected questions and the same general answer–explanation format. Their responses are also evaluated using the same procedure.

The evaluation considers more than answer accuracy. I also examine output-format compliance and generation time, and manually evaluate a subset of explanations for explanation-support faithfulness, logical consistency, completeness, and common failure types. This allows answer performance and explanation quality to be examined within the same experimental setup.

The scope of this comparison is deliberately limited to the visible model output. The aim is to determine how well a generated explanation supports the answer reported by the model. The experiments do not show whether that

explanation represents the model’s actual internal reasoning. Answer-supporting explanations and causal explanations of model behavior are therefore treated as different questions in this paper.

3 Methodology

3.1 Models and Datasets

I evaluated three compact open-weight instruction models: Qwen 2.5 3B, Llama 3.2 3B, and DeepSeek-R1 Distill Qwen 1.5B [DeepSeek-AI, 2025; Grattafiori et al., 2024; Yang et al., 2024]. Since the experiments were run locally on a GPU with 4 GB of memory, all three models were loaded using 4-bit NF4 quantization.

The evaluation used three reasoning datasets:

- **GSM8K:** Mathematical reasoning based on word problems [Cobbe et al., 2021].
- **CommonsenseQA:** Commonsense reasoning with five possible answer choices [Talmor et al., 2019].
- **StrategyQA:** Multi-hop reasoning based on yes/no questions [Geva et al., 2021].

I selected 100 questions from each dataset using random seed 42. The questions were taken from the GSM8K test split, the CommonsenseQA validation split, and the available StrategyQA training split. The same 300 selected questions were used for every model so that the models were compared on the same inputs.

3.2 Prompting and Inference

The same general prompt structure was used for all three models. Each model was asked to provide a final answer together with a short explanation. The required output format was:

```
FINAL_ANSWER: <answer>
EXPLANATION: <concise justification>
```

The expected final answer depended on the dataset. GSM8K required a numerical answer, CommonsenseQA required an option label, and StrategyQA required a yes/no answer. I placed the final answer before the explanation so that responses from all three datasets could be handled using the same general parsing procedure.

Greedy decoding was used and sampling was disabled. Qwen and Llama were allowed up to 256 new tokens for each response. I initially used the same limit for DeepSeek, but during preliminary runs some of its responses ended before both required output fields were completed. I therefore increased its final limit to 768 new tokens.

The complete experiment produced 900 responses:

$$3 \text{ models} \times 3 \text{ datasets} \times 100 \text{ questions} = 900 \text{ responses.}$$

More details about the model identifiers, hardware, and generation settings are given in Appendix A. Figure 1 gives an overview of the experimental workflow.

3.3 Automatic Evaluation

After generation, I used a strict parser to extract the final answer and the explanation from each response. I measured format compliance separately from answer correctness because a model could give the correct answer without following the requested output format.

The extracted answers were normalized according to the dataset. For GSM8K, numerical answers were normalized before comparison. CommonsenseQA responses were processed as option labels, while StrategyQA accepted equivalent yes/no, true/false, and A/B forms.

Three measures were recorded:

- **Answer accuracy:** The proportion of responses with the correct final answer.
- **Strict format compliance:** The proportion of responses containing both required fields in the expected format.
- **Generation time:** The time taken to generate each response, summarized using the mean.

I also calculated Wilson 95% confidence intervals for dataset-level and overall answer accuracy.

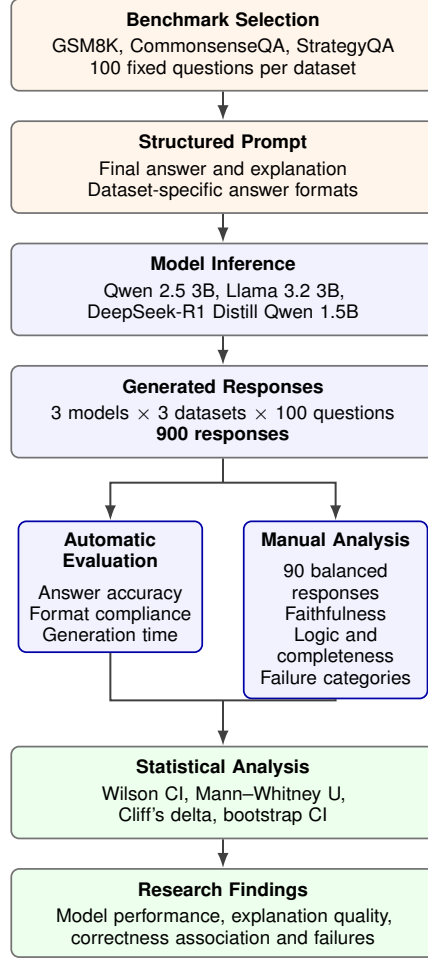


Figure 1: Experimental pipeline. Three open-weight models generated 900 responses across three reasoning benchmarks. Automatic evaluation and manual explanation analysis were followed by statistical comparison.

3.4 Manual Explanation Analysis

Automatic evaluation can show whether an answer is correct and whether the requested format was followed, but it does not tell us whether the explanation actually provides good support for the answer. I therefore selected a smaller sample of 90 successfully parsed responses for manual evaluation.

The sample contained 10 responses from every model–dataset combination. Where possible, I selected five correct and five incorrect answers for each combination. This gave equal representation to the three models and datasets and allowed explanations for correct and incorrect answers to be compared.

This sampling choice also has an important limitation. The sample was deliberately balanced by answer correctness, so the manual scores should not be interpreted as estimates of explanation quality across all 900 responses.

I scored each selected explanation from 0 to 3 on three criteria:

- **Explanation-support faithfulness:** Whether the explanation provides correct and relevant support for the reported final answer.
- **Logical consistency:** Whether the explanation follows a coherent line of reasoning without contradictions.
- **Completeness:** Whether enough information and reasoning are provided to justify the answer.

I also recorded one primary failure category where applicable. The categories included answer–explanation mismatch, arithmetic error, contradictory explanation, factual hallucination, incomplete reasoning, instruction-following failure,

logical error, unsupported justification, and other errors. Responses for which no main failure was identified were recorded separately.

The full scoring rubric and definitions of these categories are given in Appendix C.

This manual evaluation is limited to the explanation visible in the generated response. It cannot determine whether the explanation reflects the internal computations that actually produced the model’s answer. Testing that stronger form of faithfulness would require other methods, such as interventions on model behavior [Lanham et al., 2023; Parcalabescu and Frank, 2024; Turpin et al., 2023].

3.5 Statistical Analysis

I used statistical analysis to examine the accuracy results and the manually assigned faithfulness scores. Wilson 95% confidence intervals were calculated for the observed accuracy proportions.

For the manual sample, I compared faithfulness scores for correct and incorrect answers using a two-sided Mann–Whitney U test. This non-parametric test was used because the faithfulness scores were measured on an ordinal 0–3 scale. I also calculated Cliff’s delta to describe the size and direction of the difference between the two groups.

In addition, I used 10,000 bootstrap resamples with random seed 42 to calculate a 95% confidence interval for the difference in mean faithfulness scores. The comparison was first carried out using all 90 manually reviewed responses and was then repeated separately for each model.

These results describe an association between answer correctness and explanation-support faithfulness. Since correctness also contributes to the faithfulness assessment, the relationship should not be interpreted as independent evidence that the explanations reflect the models’ internal causal reasoning.

The complete model-level statistical results are reported in Appendix E.

3.6 Code and Reproducibility

The code used for data preparation, model inference, answer evaluation, manual-analysis support, statistical analysis, and figure generation is included in the project repository.

The repository is available at: github.com/Yash-Gavade/FaithfulLLM

The repository also contains the experiment configuration and supporting files used in the evaluation. Processed benchmark metadata and generated outputs are included where licensing and storage constraints permit. Further details about the experimental setup and reproduction workflow are provided in Appendix H.

4 Results

4.1 Accuracy and Efficiency

I first compared the models in terms of answer accuracy, format compliance, and generation time. The main results are shown in Table 1.

Table 1: Answer accuracy, strict-format compliance, and mean generation time.

Model	GSM (%)	CSQA (%)	SQA (%)	Overall (%)	Parse (%)	Time (s)
Qwen 2.5 3B	13	76	60	49.67	99.33	10.21
Llama 3.2 3B	6	70	59	45.00	100.00	7.57
DeepSeek-R1 Distill	80	28	49	52.33	85.67	37.01

There are large differences between the models when the three datasets are considered separately. On GSM8K, DeepSeek-R1 Distill achieved 80% accuracy, compared with 13% for Qwen and 6% for Llama. The result was very different on CommonsenseQA, where Qwen reached 76% and Llama 70%, while DeepSeek reached 28%. The three models were closer on StrategyQA, with accuracies of 60% for Qwen, 59% for Llama, and 49% for DeepSeek.

Despite these large differences on individual datasets, the overall accuracies were relatively close. DeepSeek had an overall accuracy of 52.33%, followed by Qwen with 49.67% and Llama with 45.00%. The model with the highest accuracy therefore changed depending on the reasoning task.

There were also clear differences in format compliance and generation time. Llama followed the required output format in all 300 responses and had the shortest mean generation time at 7.57 seconds. Qwen reached 99.33% format compliance and took 10.21 seconds on average. DeepSeek had lower format compliance at 85.67% and a considerably higher mean generation time of 37.01 seconds.

The timing results should be interpreted with some caution. DeepSeek was allowed a larger generation budget because shorter preliminary runs often ended before the required output was complete. The reported times therefore describe the models under the settings used in this experiment rather than comparing their speed under identical token limits.

Figure 2 shows the dataset-level accuracy results together with Wilson 95% confidence intervals.

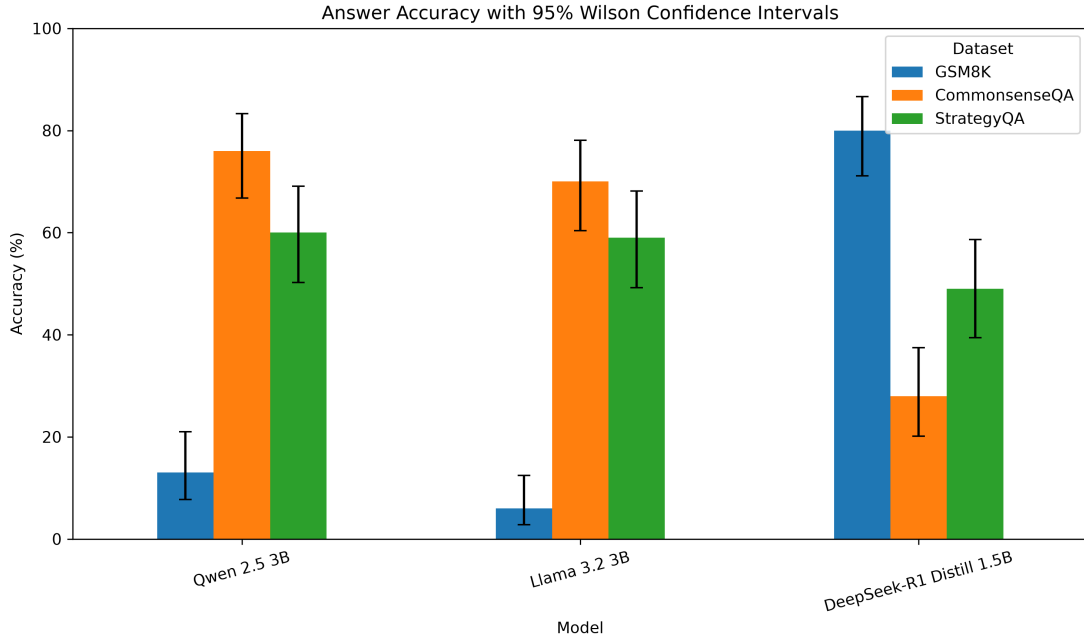


Figure 2: Answer accuracy by model and dataset with Wilson 95% confidence intervals. Each dataset-level estimate is based on 100 samples.

The task-specific differences are easier to see in Figure 2. DeepSeek performed much better than the other two models on GSM8K, but the same pattern was not observed on CommonsenseQA or StrategyQA. Qwen and Llama, in contrast, performed much better on these two datasets than on GSM8K.

The confidence intervals give some additional context to the observed differences. The largest gaps appear at the dataset level, while the overall accuracy values are closer and their confidence intervals overlap. The complete dataset-level and overall confidence intervals are reported in Appendix D.1.

4.2 Explanation Quality

Answer accuracy only describes one part of the model outputs. I also manually examined the explanations to see how well they supported the reported answers. Table 2 shows the mean scores for explanation-support faithfulness, logical consistency, and completeness. Each model contributed 30 responses to the manually reviewed sample.

Table 2: Mean reviewed explanation scores on a 0–3 scale.

Model	Faithfulness	Logic	Completeness
DeepSeek-R1 Distill	1.87	2.07	2.03
Llama 3.2 3B	1.70	2.20	2.17
Qwen 2.5 3B	1.73	2.13	2.30

No model had the highest mean score on all three criteria. DeepSeek had the highest mean explanation-support faithfulness score at 1.87. Llama had the highest logical-consistency score at 2.20, while Qwen had the highest completeness score at 2.30.

The differences in the mean faithfulness scores were relatively small: they ranged from 1.70 for Llama to 1.87 for DeepSeek on the 0–3 scale. For this reason, I treat faithfulness, logical consistency, and completeness as separate aspects of the explanations rather than combining them into a single overall score.

The scores also varied across datasets. For example, DeepSeek received higher faithfulness scores on GSM8K and CommonsenseQA than on StrategyQA within the reviewed sample. A complete breakdown by model and dataset is shown in the heatmap in Appendix F.

4.3 Correctness and Faithfulness

A larger difference appeared when I separated the manually reviewed responses into correct and incorrect answers. Figure 3 shows the mean explanation-support faithfulness score for the two groups.

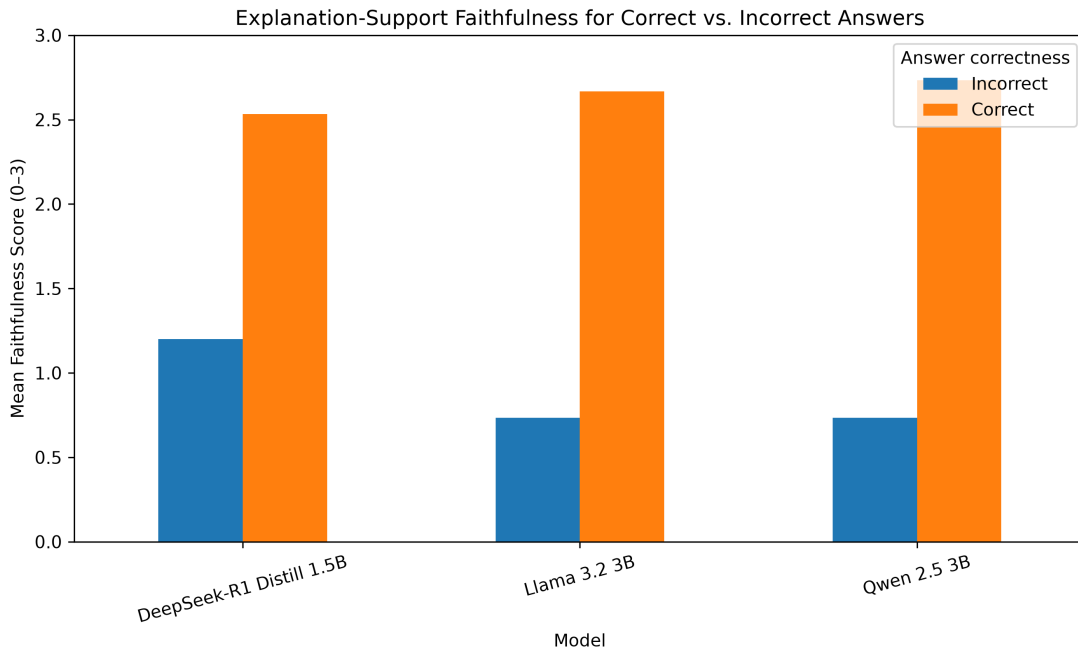


Figure 3: Mean explanation-support faithfulness for correct and incorrect answers.

Across the manually reviewed responses, explanations for correct answers had a mean faithfulness score of 2.64. For incorrect answers, the mean was 0.89. This gives a difference of 1.76 points, with a bootstrap 95% confidence interval of [1.42, 2.07].

The two groups were also compared using a two-sided Mann–Whitney U test. The difference was statistically significant ($U = 1827.0$, $p < 0.001$), and Cliff’s delta was 0.804. The same direction of difference was observed when each model was examined separately. The complete model-level results are given in Appendix E.

This result is consistent with H2. In the manually reviewed sample, correct answers were accompanied by explanations with higher explanation-support faithfulness scores. However, correctness is itself part of judging whether an explanation provides valid support for an answer. The result should therefore be understood as an association within the scoring framework used here, not as evidence that the written explanations represent the models’ internal reasoning processes.

4.4 Failure Analysis

The manual review also showed several types of explanation failure that are not captured by the mean scores alone. These included unsupported justifications, factual hallucinations, logical and arithmetic errors, contradictory explanations, and mismatches between the explanation and the final answer.

Some of these problems were easy to identify. For example, there were responses in which the reasoning produced one numerical result but the final answer reported another value. Other cases were less obvious. A model could choose an incorrect CommonsenseQA option while still giving a fluent explanation that initially appeared reasonable.

These examples show why I did not judge explanation quality only from how fluent or detailed a response appeared. The explanation was evaluated together with the reported answer and the reference answer so that unsupported or inconsistent reasoning could also be identified.

The complete distribution of failure categories is shown in Appendix F, and selected examples from the manual review are discussed in Appendix G.

5 Discussion

5.1 Task-Specific Performance

The results show that the three models performed very differently depending on the reasoning task. DeepSeek-R1 Distill performed much better on GSM8K, while Qwen achieved the highest accuracy on CommonsenseQA and StrategyQA. Llama was close to Qwen on these two datasets but performed poorly on GSM8K. This supports H1, as the performance of the models was not consistent across the three tasks.

This difference is less visible when only overall accuracy is considered. DeepSeek had the highest overall accuracy, but much of this came from its strong result on GSM8K. The same advantage was not present on CommonsenseQA or StrategyQA. Qwen and Llama showed the opposite pattern, with much better results on the two non-mathematical tasks. For this reason, overall accuracy alone does not fully describe how the models performed. Similar differences across reasoning and knowledge tasks have also been reported in broader language-model evaluations [BIG-Bench Authors, 2023; Hendrycks et al., 2021; Liang et al., 2023; Zellers et al., 2019].

The models also differed in format compliance and generation time. Llama followed the requested output format in every evaluated response and had the shortest mean generation time. Qwen also followed the format in almost all cases. DeepSeek had more formatting failures and took considerably longer to generate its responses. Its strong performance on GSM8K therefore came with a higher runtime and lower format compliance in this experiment.

However, the generation times are not directly comparable under identical conditions. DeepSeek was allowed more output tokens because shorter preliminary runs often ended before both required fields were completed. Its longer generation time is therefore partly related to the configuration used in the experiment and should not be interpreted as a controlled speed comparison.

Taken together, these results show why it is useful to consider the individual tasks and practical measures instead of looking only at one overall accuracy value. None of the three models had the same advantage across every task and measure considered in this study.

5.2 Faithfulness and Answer Correctness

The manual evaluation showed a clear difference between explanations for correct and incorrect answers. Correct answers had a mean explanation-support faithfulness score of 2.64, compared with 0.89 for incorrect answers. The same direction of difference appeared for all three models. This result is consistent with H2.

This finding needs to be interpreted carefully because answer correctness is related to the faithfulness score used in this study. If an explanation contains incorrect reasoning in support of an incorrect answer, it will normally receive a lower score. The difference between correct and incorrect responses therefore does not independently show that correct answers were produced through more faithful internal reasoning.

The manually reviewed examples also show that correctness and explanation quality do not always match. Some correct answers had incomplete explanations, while some incorrect answers were accompanied by fluent and reasonable-sounding justifications. There were also cases where the reasoning and the final answer did not agree with each other. This means that looking only at the final answer or only at the fluency of the explanation can hide important problems.

Similar concerns have been reported in previous studies, where generated reasoning appeared plausible without reliably reflecting the factors that influenced the model’s prediction [Lanham et al., 2023; Madsen et al., 2024; Matton et al., 2025; Paul et al., 2024; Turpin et al., 2023]. A detailed explanation can therefore make a response easier to inspect, but the amount of detail does not by itself make the explanation faithful.

The failure categories found during manual review give another view of this problem. The responses included unsupported justifications, factual hallucinations, contradictory reasoning, arithmetic and logical errors, and mismatches

between the explanation and the final answer. These are not all the same type of failure. For example, an explanation may be incomplete without being contradictory, while another may be fluent but depend on an unsupported claim. Looking at several aspects of explanation quality is therefore more useful than relying on one score alone. Related research on hallucination, uncertainty, and generated reasoning has identified similar reliability problems [Farquhar et al., 2024; Manakul et al., 2023; Siegel et al., 2024; Terentowicz et al., 2025].

5.3 Implications for Trustworthy XAI

An important point in this study is the difference between explanation-support faithfulness and causal faithfulness. The manual evaluation checks whether the written explanation gives relevant and consistent support for the model’s final answer. This can be evaluated from the generated response itself.

It does not tell us whether the explanation describes the internal process that actually caused the model to produce the answer. A model can generate a reasonable explanation without that explanation being an accurate description of its internal reasoning. The results in this paper should therefore be understood as an evaluation of the visible explanation, not as a direct measurement of the model’s internal reasoning process.

Mechanistic interpretability studies a different part of this problem by examining internal model components, learned features, and computational mechanisms [Bricken et al., 2023; Elhage et al., 2021; Huben et al., 2024; Rai et al., 2024]. I did not perform this type of analysis in the present study. It would require additional evidence before stronger claims about internal or causal faithfulness could be made.

This limitation does not make self-generated explanations useless. Requiring a model to explain its answer can make some problems easier to notice. For example, the explanation may reveal a contradiction, an unsupported statement, a missing reasoning step, or a mismatch with the final answer. These problems may be harder to identify when only the final answer is available.

However, an explanation should still be treated as additional information rather than proof that an answer is correct. For applications where reliable reasoning is important, explanations could be used together with other checks, such as independent answer verification, uncertainty estimation, hallucination detection, or consistency checking. Previous work has explored several of these approaches [Desai and Durrett, 2020; Farquhar et al., 2024; Manakul et al., 2023; Sui et al., 2025; Weng et al., 2023].

For the models evaluated here, the main result is that answer accuracy and explanation quality need to be considered separately. The best performing model can change with the task, and a well-written explanation can still contain incorrect or unsupported reasoning. Self-generated explanations can help us inspect a model’s response, but they should not be treated as evidence that the model has revealed the internal reasoning that produced its answer.

6 Limitations and Future Work

There are several limitations to this study. First, I evaluated only three relatively small open-weight models and used 100 questions from each dataset. Using the same questions for all three models made the comparison consistent, but the results may be different for larger models, other model families, or different reasoning tasks.

The manual evaluation was also limited to 90 of the 900 generated responses. The sample was balanced across models, datasets, and answer correctness so that these groups could be compared more easily. However, because the sample was deliberately constructed in this way, the average manual scores should not be treated as estimates of explanation quality across all 900 responses.

Another limitation is that the manual evaluation was completed using a single annotation process. I used the same scoring rubric for all reviewed responses to make the assessment as consistent as possible, but judging faithfulness, logical consistency, and completeness still involves some subjectivity. Using multiple independent annotators in future work would allow inter-annotator agreement to be measured and would provide a stronger check on the reliability of the manual scores.

The meaning of faithfulness in this paper is also intentionally limited. I evaluated whether the written explanation provides relevant and consistent support for the final answer. This does not show whether the explanation reflects the internal reasoning process that actually produced the answer [Lanham et al., 2023; Zaman and Srivastava, 2025]. Answer correctness is also related to the explanation-support score used in this study. Therefore, the observed difference between correct and incorrect answers should not be interpreted as an independent measure of causal faithfulness.

The available hardware also influenced the experimental setup. All three models were loaded using 4-bit NF4 quantization because the available GPU had 4 GB of memory. Qwen and Llama were evaluated with a limit of 256

new tokens, while DeepSeek was given 768 because shorter preliminary runs often ended before the required output was complete. For this reason, the generation times are not a controlled comparison under identical token limits. The experiment also used one standardized prompt and greedy decoding. Different prompts, decoding methods, or quantization settings could produce different results.

StrategyQA introduces another limitation. I used the available training split rather than an untouched test split. Because of this, possible training-data contamination cannot be ruled out. The StrategyQA results should therefore be interpreted more cautiously than results from an unseen evaluation set.

Future work could extend the experiment to larger models, additional model families, and more reasoning benchmarks. It would also be useful to test different prompts and decoding strategies instead of relying on one fixed setup. A larger manually reviewed sample together with multiple independent annotators would provide a stronger evaluation of explanation quality and would make it possible to measure inter-annotator agreement.

Finally, future work could examine faithfulness beyond the visible explanation. For example, causal interventions could test whether changing or removing parts of the generated reasoning changes the model’s prediction [Matton et al., 2025; Zaman and Srivastava, 2025]. Combining this type of analysis with consistency checks and other reliability measures could help distinguish explanations that only appear convincing from those that provide stronger evidence about the model’s behavior.

7 Conclusion

In this study, I evaluated self-generated explanations from three open-weight LLMs: Qwen 2.5 3B, Llama 3.2 3B, and DeepSeek-R1 Distill Qwen 1.5B. The models were tested on GSM8K, CommonsenseQA, and StrategyQA, giving a total of 900 responses. I also manually reviewed a balanced sample of 90 responses to examine explanation-support faithfulness, logical consistency, completeness, and common explanation failures.

The results showed that the models had different strengths across the three tasks. DeepSeek achieved the highest accuracy on GSM8K, while Qwen performed best on CommonsenseQA and StrategyQA. Llama had the shortest mean generation time in the experimental setup and followed the required output format in all 300 responses. These results show that overall accuracy alone does not give a complete picture of model performance.

The manual evaluation showed an even clearer difference between correct and incorrect answers. Explanations for correct answers had a mean faithfulness score of 2.64, compared with 0.89 for incorrect answers. However, the manual review also found problems in the explanations from all three models. These included unsupported claims, incomplete or incorrect reasoning, and cases where the explanation did not agree with the final answer. A fluent and convincing explanation is therefore not necessarily a reliable explanation.

The results should also be understood within the definition of faithfulness used in this paper. I evaluated whether the visible explanation provides adequate support for the answer reported by the model. This does not show whether the explanation represents the internal reasoning process that actually produced the answer. Making that stronger claim would require a different type of evaluation.

Overall, self-generated explanations are useful because they provide more information about a model’s response than the final answer alone. They can make reasoning errors, unsupported statements, and inconsistencies easier to identify. However, they should be treated as additional evidence rather than proof that a model’s reasoning is correct or faithful. For applications where reliable reasoning is important, the answer and its explanation should still be checked using additional forms of verification.

References

- BIG-Bench Authors [2023]. “Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models”. In: *Transactions on Machine Learning Research*. Code or resources: <https://github.com/google/BIG-bench>. URL: <https://openreview.net/forum?id=uyTL5Bvosj>.
- Bricken, T. et al. [2023]. *Towards Monosemanticity: Decomposing Language Models With Dictionary Learning*. Anthropic Research. URL: <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- Cobbe, K. et al. [2021]. “Training Verifiers to Solve Math Word Problems”. In: *arXiv preprint arXiv:2110.14168*. Code or resources: <https://github.com/openai/grade-school-math>. arXiv: 2110.14168. URL: <https://arxiv.org/abs/2110.14168>.
- DeepSeek-AI [2025]. *DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning*. Technical Report. Code or resources: <https://github.com/deepseek-ai/DeepSeek-R1>. DeepSeek-AI. arXiv: 2501.12948. URL: <https://arxiv.org/abs/2501.12948>.
- Desai, S. and Durrett, G. [2020]. “Calibration of Pre-trained Transformers”. In: *EMNLP*. DOI: 10.18653/v1/2020.emnlp-main.21. URL: <https://aclanthology.org/2020.emnlp-main.21/>.
- Elhage, N. et al. [2021]. *A Mathematical Framework for Transformer Circuits*. Transformer Circuits. URL: <https://transformer-circuits.pub/2021/framework/index.html>.
- Farquhar, S. et al. [2024]. “Detecting Hallucinations in Large Language Models Using Semantic Entropy”. In: *Nature*. Code: https://github.com/jlko/semantic_uncertainty. DOI: 10.1038/s41586-024-07421-0. URL: <https://www.nature.com/articles/s41586-024-07421-0>.
- Geva, M. et al. [2021]. “Did Aristotle Use a Laptop? A Question Answering Benchmark with Implicit Reasoning Strategies”. In: *TACL*. Code or resources: <https://github.com/eladsegal/strategyqa>. DOI: 10.1162/tacl_a_00370. URL: <https://aclanthology.org/2021.tacl-1.21/>.
- Grattafiori, A. et al. [2024]. *The Llama 3 Herd of Models*. Technical Report. Code or resources: <https://github.com/meta-llama/llama-models>. Meta. arXiv: 2407.21783. URL: <https://arxiv.org/abs/2407.21783>.
- Hendrycks, D. et al. [2021]. “Measuring Massive Multitask Language Understanding”. In: *ICLR*. Code or resources: <https://github.com/hendrycks/test>. URL: <https://openreview.net/forum?id=d7KBjmI3GmQ>.
- Huben, R., Cunningham, H., Smith, L. R., Ewart, A., and Sharkey, L. [2024]. “Sparse Autoencoders Find Highly Interpretable Features in Language Models”. In: *International Conference on Learning Representations*. Code: https://github.com/HoagyC/sparse_coding. URL: <https://openreview.net/forum?id=F76bwRSLeK>.
- Lanham, T. et al. [2023]. “Measuring Faithfulness in Chain-of-Thought Reasoning”. In: *arXiv preprint arXiv:2307.13702*. arXiv: 2307.13702. URL: <https://arxiv.org/abs/2307.13702>.
- Liang, P. et al. [2023]. “Holistic Evaluation of Language Models”. In: *Transactions on Machine Learning Research*. Code or resources: <https://github.com/stanford-crfm/helm>. URL: <https://openreview.net/forum?id=i04LZibEqW>.
- Lipton, Z. C. [2018]. “The Mythos of Model Interpretability”. In: *Communications of the ACM*. DOI: 10.1145/3233231. URL: <https://dl.acm.org/doi/10.1145/3233231>.
- Lundberg, S. M. and Lee, S.-I. [2017]. “A Unified Approach to Interpreting Model Predictions”. In: *NeurIPS*. Code or resources: <https://github.com/shap/shap>. URL: <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>.
- Madaan, A. et al. [2023]. “Self-Refine: Iterative Refinement with Self-Feedback”. In: *NeurIPS*. Code or resources: <https://github.com/madaan/self-refine>. URL: https://proceedings.neurips.cc/paper_files/paper/2023/hash/91edff07232fb1b55a505a9e9f6c0ff3-Abstract-Conference.html.
- Madsen, A., Chandar, S., and Reddy, S. [2024]. “Are Self-Explanations from Large Language Models Faithful?” In: *Findings of ACL*. Code or resources: <https://github.com/AndreasMadsen/llm-self-explanations>. DOI: 10.18653/v1/2024.findings-acl.19. URL: <https://aclanthology.org/2024.findings-acl.19/>.
- Manakul, P., Liusie, A., and Gales, M. [2023]. “SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models”. In: *EMNLP*. Code or resources: <https://github.com/potsawee/selfcheckgpt>. DOI: 10.18653/v1/2023.emnlp-main.557. URL: <https://aclanthology.org/2023.emnlp-main.557/>.
- Matton, K. et al. [2025]. “Walk the Talk? Measuring the Faithfulness of Large Language Model Explanations”. In: *ICLR*. URL: <https://openreview.net/forum?id=4ub9gpx9xw>.
- Miller, T. [2019]. “Explanation in Artificial Intelligence: Insights from the Social Sciences”. In: *Artificial Intelligence*. DOI: 10.1016/j.artint.2018.07.007. URL: <https://www.sciencedirect.com/science/article/pii/S0004370218305988>.
- Parcalabescu, L. and Frank, A. [2024]. “On Measuring Faithfulness or Self-Consistency of Natural Language Explanations”. In: *ACL*. DOI: 10.18653/v1/2024.acl-long.329. URL: <https://aclanthology.org/2024.acl-long.329/>.

- Paul, D. et al. [2024]. “Making Reasoning Matter: Measuring and Improving Faithfulness of Chain-of-Thought Reasoning”. In: *Findings of the Association for Computational Linguistics: EMNLP 2024*. DOI: [10.18653/v1/2024.findings-emnlp.882](https://doi.org/10.18653/v1/2024.findings-emnlp.882). URL: <https://aclanthology.org/2024.findings-emnlp.882/>.
- Rai, D. et al. [2024]. “A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models”. In: *arXiv preprint arXiv:2407.02646*. arXiv: [2407.02646](https://arxiv.org/abs/2407.02646). URL: <https://arxiv.org/abs/2407.02646>.
- Ribeiro, M. T., Singh, S., and Guestrin, C. [2016]. ““Why Should I Trust You?”: Explaining the Predictions of Any Classifier”. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Code: <https://github.com/marcotcr/lime>. ACM, pp. 1135–1144. DOI: [10.1145/2939672.2939778](https://doi.org/10.1145/2939672.2939778). URL: <https://doi.org/10.1145/2939672.2939778>.
- Schneider, J. [2024]. “Explainable Generative AI (GenXAI): A Survey, Conceptualization, and Research Agenda”. In: *Artificial Intelligence Review*. DOI: [10.1007/s10462-024-10810-x](https://doi.org/10.1007/s10462-024-10810-x). URL: <https://link.springer.com/article/10.1007/s10462-024-10810-x>.
- Siegel, N. Y. et al. [2024]. “The Probabilities Also Matter: A More Faithful Metric for Faithfulness of Free-Text Explanations”. In: *ACL*. DOI: [10.18653/v1/2024.acl-short.49](https://doi.org/10.18653/v1/2024.acl-short.49). URL: <https://aclanthology.org/2024.acl-short.49/>.
- Sui, Y. et al. [2025]. “FiDeLiS: Faithful Reasoning in Large Language Models for Knowledge-Grounded Question Answering”. In: *Findings of ACL*. DOI: [10.18653/v1/2025.findings-acl.436](https://doi.org/10.18653/v1/2025.findings-acl.436). URL: <https://aclanthology.org/2025.findings-acl.436/>.
- Talmor, A. et al. [2019]. “CommonsenseQA: A Question Answering Challenge Targeting Commonsense Knowledge”. In: *NAACL*. Code or resources: <https://www.tau-nlp.sites.tau.ac.il/commonsenseqa>. DOI: [10.18653/v1/N19-1421](https://doi.org/10.18653/v1/N19-1421). URL: <https://aclanthology.org/N19-1421/>.
- Terentowicz, A. et al. [2025]. “How (Un)faithful are Explainable LLM-based NLG Metrics?” In: *INLG*. DOI: [10.18653/v1/2025.inlg-main.37](https://doi.org/10.18653/v1/2025.inlg-main.37). URL: <https://aclanthology.org/2025.inlg-main.37/>.
- Turpin, M. et al. [2023]. “Language Models Don’t Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting”. In: *NeurIPS*. URL: https://proceedings.neurips.cc/paper_files/paper/2023/hash/ed3fea9033a80fea1376299fa7863f4a-Abstract-Conference.html.
- Wang, X. et al. [2023]. “Self-Consistency Improves Chain-of-Thought Reasoning in Language Models”. In: *ICLR*. Code or resources: <https://github.com/google-research/chain-of-thought-hub>. URL: <https://openreview.net/forum?id=1PL1NIMMrw>.
- Wei, J. et al. [2022]. “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models”. In: *NeurIPS*. Code or resources: <https://github.com/google-research/chain-of-thought-hub>. URL: https://proceedings.neurips.cc/paper_files/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html.
- Weng, Y. et al. [2023]. “Large Language Models are Better Reasoners with Self-Verification”. In: *Findings of the Association for Computational Linguistics: EMNLP 2023*. Code or resources: <https://github.com/WENGSYX/Self-Verification>. DOI: [10.18653/v1/2023.findings-emnlp.167](https://doi.org/10.18653/v1/2023.findings-emnlp.167). URL: <https://aclanthology.org/2023.findings-emnlp.167/>.
- Yang, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Wei, H., Lin, H., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Lin, J., Dang, K., Lu, K., Bao, K., Yang, K., Yu, L., Li, M., Xue, M., Zhang, P., Zhu, Q., Men, R., Lin, R., Li, T., Tang, T., Xia, T., Ren, X., Ren, X., Fan, Y., Su, Y., Zhang, Y., Wan, Y., Liu, Y., Cui, Z., Zhang, Z., and Qiu, Z. [2024]. *Qwen2.5 Technical Report*. arXiv: [2412.15115](https://arxiv.org/abs/2412.15115). URL: <https://arxiv.org/abs/2412.15115>.
- Zaman, K. and Srivastava, S. [2025]. “A Causal Lens for Evaluating Faithfulness Metrics”. In: *EMNLP*. DOI: [10.18653/v1/2025.emnlp-main.1496](https://doi.org/10.18653/v1/2025.emnlp-main.1496). URL: <https://aclanthology.org/2025.emnlp-main.1496/>.
- Zelikman, E. et al. [2022]. “STaR: Bootstrapping Reasoning With Reasoning”. In: *NeurIPS*. URL: https://proceedings.neurips.cc/paper_files/paper/2022/hash/639a9a172c044fbb64175b5fad42e9a5-Abstract-Conference.html.
- Zellers, R. et al. [2019]. “HellaSwag: Can a Machine Really Finish Your Sentence?” In: *ACL*. Code or resources: <https://github.com/rowanz/hellaswag>. DOI: [10.18653/v1/P19-1472](https://doi.org/10.18653/v1/P19-1472). URL: <https://aclanthology.org/P19-1472/>.

Appendix

A Experimental Setup

A.1 Datasets and Models

I used three open-weight instruction models in the experiments: Qwen2.5-3B-Instruct, Llama-3.2-3B-Instruct, and DeepSeek-R1-Distill-Qwen-1.5B. The models were evaluated on GSM8K, CommonsenseQA, and StrategyQA. These datasets represent mathematical reasoning, commonsense question answering, and multi-hop yes/no reasoning, respectively [Cobbe et al., 2021; Geva et al., 2021; Talmor et al., 2019].

For each dataset, I selected 100 questions using random seed 42. The same 300 questions were used for all three models so that every model was evaluated on the same inputs. This produced 900 responses in total:

$$3 \text{ models} \times 3 \text{ datasets} \times 100 \text{ questions} = 900 \text{ responses.}$$

The same general prompting and answer-processing procedure was used for all three models. The main settings are summarized in Table 3.

Table 3: Summary of the experimental setup.

Component	Configuration
Datasets	GSM8K, CommonsenseQA, StrategyQA
Dataset size	100 questions per dataset
Models	Qwen 2.5 3B, Llama 3.2 3B, DeepSeek-R1 Distill Qwen 1.5B
Total responses	900
Sampling seed	42
Quantization	4-bit NF4
Decoding	Greedy; sampling disabled

Table 3 gives the main settings used throughout the experiment. The number of questions and the general inference procedure were kept fixed across the three models. Using the same questions made the comparison more consistent. The models were run locally using 4-bit quantization because of the available GPU memory.

A.2 Hardware and Generation Settings

I ran the experiments locally on a Windows computer with an NVIDIA GeForce GTX 1650 Ti GPU and 4 GB of VRAM. The software environment included Python 3.11.2, PyTorch 2.7.1, CUDA 11.8, Hugging Face Transformers, Datasets, bitsandbytes, and Accelerate.

Because the GPU had only 4 GB of memory, I loaded all three models using 4-bit NF4 quantization with double quantization and Float16 computation. The same quantization settings were used for every model.

For generation, I used greedy decoding with sampling disabled. Qwen and Llama were allowed up to 256 new tokens per response. I initially used the same limit for DeepSeek, but during preliminary runs some responses ended before both required output fields were completed. I therefore increased the final limit for DeepSeek to 768 new tokens.

This difference needs to be considered when comparing generation time. DeepSeek had a larger generation budget than Qwen and Llama, so its longer runtime should not be interpreted as a controlled speed comparison under identical token limits. The reported times instead describe the models under the settings actually used in the experiment.

For StrategyQA, I used the available training split rather than an untouched test split. This creates a possible risk of training-data contamination that cannot be ruled out in this experiment. For this reason, the StrategyQA results should be interpreted with additional caution.

B Prompting and Output Processing

B.1 Prompt Structure

I used the same general prompt structure for all three models. Each model received a reasoning question and was asked to return a final answer together with a short explanation.

The response was required to contain two fields, `FINAL_ANSWER` and `EXPLANATION`. The general prompt template was:

```
You are given a reasoning question.
Answer the question and provide a concise explanation.
Return exactly:
FINAL_ANSWER: <answer>
EXPLANATION: <concise justification>
```

For each inference run, the question was inserted into this template. For CommonsenseQA, I also included the five answer options because the model needed to choose from the available choices.

The expected final answer depended on the dataset. GSM8K required a numerical answer, CommonsenseQA required an option label (A–E), and StrategyQA required a yes/no answer. Apart from these dataset-specific requirements, I kept the same general prompt structure for all three models. I did not use separate model-specific prompts in the final experiment.

B.2 Output Extraction and Evaluation

After generation, I processed each response to separate the reported final answer from its explanation. The extracted answer was then compared with the reference answer using the expected format for the corresponding dataset.

The normalization procedure depended on the dataset. GSM8K answers were processed as numerical values, while CommonsenseQA answers were processed as option labels. For StrategyQA, equivalent yes/no, true/false, and A/B forms were accepted. This allowed small differences in answer format to be handled without changing whether the underlying answer was considered correct.

I kept answer correctness and explanation quality separate during the evaluation. Correctness was determined automatically from the extracted final answer, while the explanation was saved for the later manual analysis. This was important because a correct answer can still be accompanied by an incomplete or incorrect explanation.

I also kept the original generated response together with the processed output. This made it possible to inspect responses where an answer was missing, could not be parsed, or did not agree with the explanation. The exact extraction and normalization procedures are implemented in the project’s evaluation scripts.

B.3 Output Validation and Parsing

Strict-format compliance was evaluated separately from answer correctness. I considered a response strictly compliant when both required fields, `FINAL_ANSWER` and `EXPLANATION`, were present in the expected structure.

The two measures were kept separate because they describe different parts of the response. A model could give the correct answer without following the required format. Similarly, a response could follow the format correctly but contain an incorrect answer. Strict-format compliance therefore measures whether the requested output structure was followed, not whether the answer itself was correct.

This was especially relevant for DeepSeek. In preliminary runs using a 256-token limit, some responses ended before both required fields were completed. I therefore increased the maximum generation length to 768 new tokens for the final DeepSeek runs.

For every response, I retained the original generated text, extracted answer, extracted explanation, and strict-parse status. Keeping these fields separately allowed answer accuracy, format compliance, and explanation quality to be analyzed independently.

C Manual Annotation Protocol

Answer correctness was evaluated automatically for all 900 responses, but this does not show whether an explanation gives adequate support for the reported answer. I therefore selected a smaller sample for manual review.

For the manual evaluation, I considered three aspects of explanation quality: explanation-support faithfulness, logical consistency, and completeness. I also recorded one primary failure category for each reviewed response where a failure was identified.

C.1 Sample Selection

The manual sample contained 90 successfully parsed responses. I selected 10 responses from each model–dataset combination, giving 30 reviewed responses for each model and 30 for each dataset.

Where possible, each model–dataset group contained five correct and five incorrect answers. I used this balanced sample so that explanations for correct and incorrect answers could be compared while keeping equal representation across models and datasets.

This means that the manual sample does not follow the natural proportion of correct and incorrect answers in the full set of 900 responses. The average manual scores should therefore be interpreted only within the reviewed sample and not as estimates for all generated responses.

C.2 Evaluation Dimensions and Rubric

I evaluated each selected explanation separately on three dimensions:

- **Explanation-support faithfulness:** whether the visible explanation provides valid support for the reported final answer.
- **Logical consistency:** whether the explanation follows coherent reasoning without invalid inferences or internal contradictions.
- **Completeness:** whether the explanation contains enough information and reasoning to justify the reported answer.

Each dimension was scored from 0 to 3. The meaning of the scores is summarized in Table 4.

Table 4: General interpretation of the manual explanation-scoring rubric.

Score	Interpretation
0	No meaningful support; the explanation is absent, unrelated, or contradictory.
1	Weak support with major factual, logical, or arithmetic problems.
2	Partial support with missing, incomplete, or flawed reasoning.
3	Clear, correct, logically consistent, and sufficient support.

I scored the three dimensions separately rather than combining them into one overall score. For example, an explanation could be logically consistent but still receive a lower completeness score because an important reasoning step was missing. Similarly, a detailed explanation could still provide weak support if its reasoning was incorrect or did not agree with the reported answer.

The three scores therefore describe different aspects of the visible explanation rather than one overall measure of explanation quality.

C.3 Primary Failure Classification

In addition to the three scores, I assigned one primary failure category to each reviewed response where a problem was identified. The purpose was to record the main problem in the response rather than every possible weakness.

The categories used in the manual review are listed in Table 5.

Table 5: Primary failure categories used during manual evaluation.

Category	Description
Answer–explanation mismatch	The explanation supports a different answer from the reported prediction.
Arithmetic error	An incorrect numerical calculation occurs in the reasoning.
Contradictory explanation	The explanation contains mutually inconsistent reasoning.
Factual hallucination	The explanation contains an unsupported or false factual claim.
Incomplete reasoning	A necessary reasoning step is missing.
Instruction-following failure	The response does not follow the required output instructions.
Logical error	The reasoning contains an invalid inference that is not primarily arithmetic.
Unsupported justification	The conclusion is stated without sufficient supporting reasoning.
Other	A failure is present but does not fit the preceding categories.
No primary failure	No major explanation failure was identified.

Only one primary category was recorded for each reviewed response. If a response contained several problems, I selected the category that best described the main observed failure. The purpose of the taxonomy was therefore to summarize the dominant problem rather than to record every possible issue in a response.

As with the numerical scores, this classification is limited to the generated text. I evaluated whether the visible explanation supported the reported answer and whether its reasoning was consistent and sufficiently complete. The annotation does not determine whether the explanation reflects the model’s internal causal reasoning.

D Detailed Experimental Results

This section gives the detailed numerical results behind the summary in the main Results section. It includes answer accuracy with confidence intervals, strict-format compliance, and mean generation time for the three models.

D.1 Answer Accuracy

I calculated answer accuracy by comparing each extracted final answer with its reference answer. Each model was evaluated on 100 questions from each dataset, giving 300 responses per model.

Table 6 gives the number of correct answers, accuracy percentages, and Wilson 95% confidence intervals for each model–dataset combination. The overall values are based on all 300 responses from each model.

Table 6: Answer accuracy and Wilson 95% confidence intervals. Accuracy and confidence-interval values are reported as percentages.

Model	Dataset	Correct	Acc. (%)	95% CI
Qwen	GSM8K	13/100	13.00	[7.76, 20.98]
	CSQA	76/100	76.00	[66.77, 83.31]
	StrategyQA	60/100	60.00	[50.20, 69.06]
	Overall	149/300	49.67	[44.05, 55.29]
Llama	GSM8K	6/100	6.00	[2.78, 12.48]
	CSQA	70/100	70.00	[60.42, 78.11]
	StrategyQA	59/100	59.00	[49.20, 68.13]
	Overall	135/300	45.00	[39.47, 50.66]
DeepSeek	GSM8K	80/100	80.00	[71.12, 86.66]
	CSQA	28/100	28.00	[20.14, 37.49]
	StrategyQA	49/100	49.00	[39.42, 58.65]
	Overall	157/300	52.33	[46.69, 57.92]

Note: CSQA denotes CommonsenseQA. Qwen, Llama, and DeepSeek refer to the model versions specified in Appendix A.

The largest accuracy differences appeared at the dataset level. DeepSeek achieved 80% on GSM8K, compared with 13% for Qwen and 6% for Llama. On CommonsenseQA, Qwen reached 76% and Llama 70%, while DeepSeek reached 28%. The StrategyQA results were closer, with Qwen at 60%, Llama at 59%, and DeepSeek at 49%.

Across all three datasets, the overall accuracies were 49.67% for Qwen, 45.00% for Llama, and 52.33% for DeepSeek. The Wilson confidence intervals for these overall values overlap. The overall results should therefore be considered together with the much larger differences found on the individual datasets.

The Wilson intervals describe uncertainty around the observed accuracy estimates under a binomial model. They do not represent variation across repeated model runs because each input was evaluated once using the same deterministic generation settings.

D.2 Output Compliance and Generation Time

In addition to correctness, I recorded whether each response followed the required output format and how long generation took. A response satisfied the strict-format requirement when the expected FINAL_ANSWER and EXPLANATION structure could be successfully parsed.

Table 7 gives the strict-format compliance and mean generation time for each model.

Table 7: Strict-format compliance and observed mean generation time.

Model	Strict parse	Mean time (s)
Qwen	298/300 (99.33%)	10.21
Llama	300/300 (100.00%)	7.57
DeepSeek	257/300 (85.67%)	37.01

Llama followed the required format in all 300 responses. Qwen had 298 strictly parsed responses, giving 99.33% compliance, while DeepSeek had 257, giving 85.67%. This corresponds to two strict-format failures for Qwen and 43 for DeepSeek.

Llama also had the shortest observed mean generation time at 7.57 seconds per response. Qwen required 10.21 seconds on average, while DeepSeek required 37.01 seconds.

These times are not a controlled comparison under identical generation budgets. Qwen and Llama were allowed up to 256 new tokens, while DeepSeek was allowed up to 768 because some shorter preliminary responses ended before the required fields were complete. The reported times therefore describe the models under the configurations actually used in the experiment.

Strict-format compliance and answer correctness were also treated as separate measures. A response could follow the requested structure and still contain an incorrect answer. It was also possible for a response to contain the correct answer without satisfying the strict parsing requirements. Keeping the two measures separate makes it possible to distinguish task performance from output-format compliance.

E Statistical Analysis

I used the 90 manually reviewed responses described in Appendix C to examine the relationship between answer correctness and explanation-support faithfulness.

E.1 Statistical Methods

I divided the reviewed responses into two groups according to whether their final answers were correct or incorrect. I then used three methods to compare their explanation-support faithfulness scores:

- **Mann–Whitney U test:** a two-sided non-parametric test used to compare the distributions of the ordinal faithfulness scores for correct and incorrect answers.
- **Cliff’s delta:** an effect-size measure used to describe the direction and magnitude of the difference between the two groups.
- **Bootstrap confidence interval:** a non-parametric bootstrap procedure used to estimate a 95% confidence interval for the difference in mean faithfulness scores between the two groups.

For the bootstrap procedure, I used 10,000 resamples with random seed 42. The main analysis combined responses from all three models and datasets. I also repeated the same comparison separately for each model.

E.2 Statistical Results

Table 8 gives the overall comparison. It includes the group means, their difference, the bootstrap confidence interval, the Mann–Whitney U statistic, and Cliff’s delta.

Table 8: Overall comparison of explanation-support faithfulness between correct and incorrect answers.

Measure	Result
Mean score: correct answers	2.64
Mean score: incorrect answers	0.89
Mean difference	1.76
Bootstrap 95% CI	[1.42, 2.07]
Mann–Whitney U	1827.0
p -value	< 0.001
Cliff’s δ	0.804

Explanations accompanying correct answers had a mean faithfulness score of 2.64, compared with 0.89 for incorrect answers. Using the unrounded group means, the difference was approximately 1.76 points. The bootstrap 95% confidence interval for this difference was [1.42, 2.07].

The two-sided Mann–Whitney U test found a difference between the two groups ($U = 1827.0$, $p < 0.001$). Cliff’s delta was 0.804, with higher faithfulness scores generally associated with correct answers in the reviewed sample.

I also repeated the analysis separately for Qwen, Llama, and DeepSeek to check whether the same direction of difference appeared within each model. The results are given in Table 9.

Table 9: Model-specific comparisons of explanation-support faithfulness.

Model	Correct	Incorrect	U	δ
Qwen	2.73	0.73	214.0	0.902
Llama	2.67	0.73	211.5	0.880
DeepSeek	2.53	1.20	183.0	0.627

The same direction of difference appeared for all three models. For Qwen, the mean faithfulness scores were 2.73 for correct answers and 0.73 for incorrect answers. The corresponding values were 2.67 and 0.73 for Llama, and 2.53 and 1.20 for DeepSeek.

The bootstrap 95% confidence intervals for the mean differences were [1.47, 2.47] for Qwen, [1.40, 2.40] for Llama, and [0.67, 1.93] for DeepSeek. The Mann–Whitney tests gave $p < 0.001$ for Qwen and Llama and $p = 0.001$ for DeepSeek. Cliff’s delta was positive for all three models, with values of 0.902, 0.880, and 0.627, respectively.

These results show that the overall pattern was not caused by only one model. Within the reviewed sample for each model, explanations for correct answers received higher faithfulness scores than explanations for incorrect answers.

E.3 Interpretation and Limitations

The statistical results show an association between answer correctness and explanation-support faithfulness in the manually reviewed sample. There are, however, several limits to how this result can be interpreted.

First, answer correctness is related to the definition of explanation-support faithfulness used in this study. The rubric considers whether an explanation gives valid support for the reported answer. Correctness and faithfulness are therefore not completely independent in this evaluation.

Second, the analysis uses a deliberately balanced sample of 90 responses rather than all 900 generated responses. I selected this sample to compare correct and incorrect answers while keeping representation across models and datasets balanced. The reported means and statistical results should therefore not be treated as estimates of the natural distribution of faithfulness scores across all model outputs.

Third, the statistical tests describe relationships in the observed responses. I did not perform causal interventions on the models or their generated reasoning. These results therefore cannot show whether the visible explanations represent the internal computations that produced the answers.

Finally, the model-specific analyses compare correct and incorrect responses within each model. They do not test whether one model has significantly higher faithfulness than another. A separate between-model analysis would be needed to make that comparison.

F Additional Figures and Failure Analysis

This section gives additional results from the manual review of 90 explanations. I first compare explanation-support faithfulness across models and datasets and then examine the primary failure categories found during annotation.

F.1 Faithfulness Across Models and Datasets

The faithfulness scores varied across both models and reasoning tasks. Figure 4 shows the mean score for each model–dataset combination. Each cell contains 10 manually reviewed responses, and the scores range from 0 to 3.

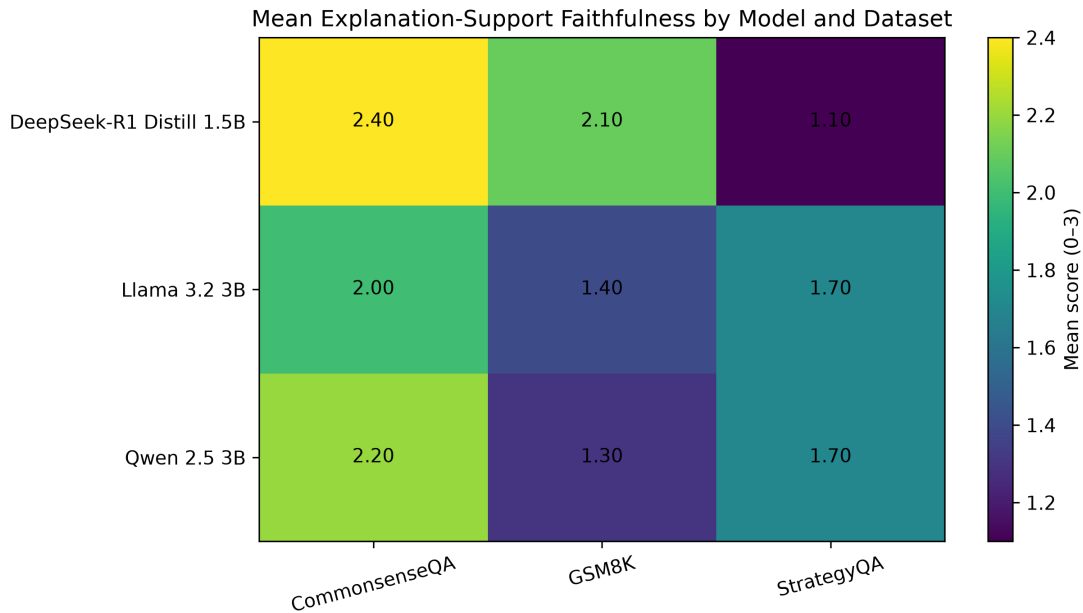


Figure 4: Mean explanation-support faithfulness across models and reasoning datasets. Scores range from 0 to 3, and each model–dataset combination contains ten manually reviewed responses.

DeepSeek received mean faithfulness scores of 2.10 on GSM8K, 2.40 on CommonsenseQA, and 1.10 on StrategyQA. Qwen received scores of 1.30, 2.20, and 1.70, respectively, while Llama received 1.40, 2.00, and 1.70. In this manually reviewed sample, CommonsenseQA had the highest mean faithfulness score for all three models.

The heatmap also shows that an overall model average can hide differences between reasoning tasks. For example, a model can have a relatively high average score while receiving a much lower score on one dataset. However, each cell contains only 10 reviewed responses. These values should therefore be treated as descriptive results from the manual sample rather than precise estimates of each model’s general performance.

F.2 Primary Explanation Failures

During the manual review, I also recorded one primary failure category for each response where a problem was identified. Figure 5 shows how these categories were distributed across the three models. Each model contributed 30 responses to the manual sample.

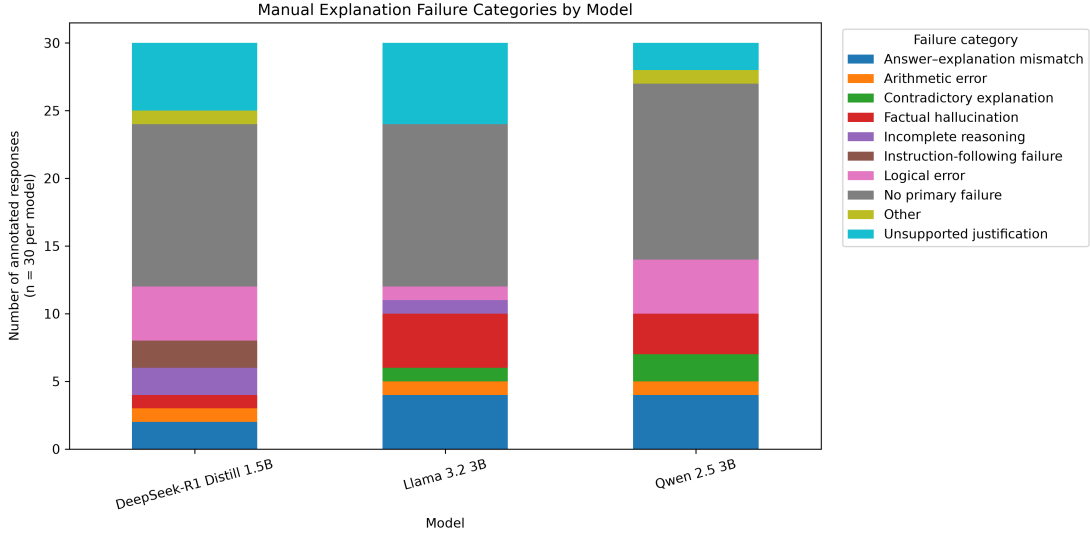


Figure 5: Distribution of primary explanation failure categories across the three evaluated models. Each model contributed 30 manually reviewed responses.

The reviewed explanations did not fail in only one way. Unsupported justifications appeared among the reviewed DeepSeek and Llama responses. Factual hallucinations and answer–explanation mismatches appeared among the Llama and Qwen responses, while contradictory explanations were also found among Qwen’s reviewed responses.

These categories describe different problems. An answer–explanation mismatch means that the explanation supports a conclusion different from the reported final answer. An unsupported justification, in contrast, means that the explanation does not provide enough support for the stated conclusion. The complete definitions are given in Appendix C.3.

This analysis captures problems that answer accuracy alone cannot show. A correct final answer can still have an incomplete or problematic explanation, while an incorrect answer can be accompanied by fluent reasoning. Looking at the failure categories therefore gives additional information about how the explanations can break down.

These results are limited to the manually reviewed sample. The figures summarize 90 deliberately balanced responses rather than all 900 model outputs. They should therefore not be interpreted as estimates of how often these failures occur across the complete experiment. The categories also describe only problems visible in the generated text and do not show whether the explanations reflect the models’ internal causal reasoning.

G Qualitative Case Studies

This section gives three examples from the manually reviewed responses. They illustrate different relationships between the reported answer and the visible explanation. The excerpts are shortened for readability.

G.1 Case 1: Correct and Supported Answer

Model: DeepSeek-R1-Distill-Qwen-1.5B **Dataset:** CommonsenseQA

Reference: E. Tiredness **Prediction:** E

Explanation excerpt:

Tiredness is likely because running after a ball requires exerting energy, leading to fatigue.

Primary failure: None.

The model selected the correct option and connected physical exertion with tiredness. The explanation therefore provides clear support for the reported answer.

G.2 Case 2: Unsupported Justification

Model: Qwen2.5-3B-Instruct **Dataset:** CommonsenseQA

Reference: D. Outside **Prediction:** C. Windowsill

Explanation excerpt:

Ivy plants often require sunlight, which they can receive from warm south- or west-facing windowsills. Windowsills also provide a stable surface.

Primary failure: Unsupported justification.

The explanation sounds reasonable, but the predicted option does not match the reference answer and the reasoning does not provide enough support for treating the selected option as the intended answer. This shows why a plausible explanation is not necessarily a reliable one.

G.3 Case 3: Mathematical Answer Mismatch

Model: Llama-3.2-3B-Instruct **Dataset:** GSM8K

Reference: 720 **Prediction:** 960

Explanation excerpt:

Greta first puts 50% of her \$2400 paycheck into retirement, which is \$1200. She then uses 20% of the paycheck for her car payment, which is \$480.

Primary failure: Answer–explanation mismatch.

The model reported 960, while the reference answer was 720. The excerpt correctly calculates \$1200 and \$480, but these calculations do not establish the reported answer of 960. I inspected the complete response during manual annotation, where it was recorded as an answer–explanation mismatch.

G.4 Overall Observation

The three cases illustrate a correct answer with adequate support, an incorrect answer with a plausible but unsupported justification, and a mathematical response where the explanation does not establish the final answer. They are illustrative examples rather than estimates of how frequently these behaviors occur across all 900 responses.

As with the rest of the manual evaluation, these examples concern only the visible model output and do not show whether the explanations reflect the models’ internal causal reasoning.

H Reproducibility and Research Contribution

H.1 Experimental Data and Records

The final benchmark contained 300 selected questions: 100 each from GSM8K, CommonsenseQA, and StrategyQA. I used random seed 42 for the sample selection and stored the processed benchmark in both CSV and JSONL formats.

The SHA-256 checksum of the processed benchmark CSV file was:

35b721529e523d4a945407053ded18d93489e5476a23ec90fad1923fb36906e5

The same 300 questions were used for all three models, producing 900 responses in total. For each response, I retained the model and dataset identifiers, input prompt, original generated text, extracted answer and explanation, reference answer, correctness label, generation time, strict-parse status, and any processing error.

The responses were saved incrementally in JSONL format. This allowed an interrupted inference run to continue without generating completed responses again. I also kept the original model outputs so that parsing problems could be inspected and the complete responses could be checked again during manual analysis.

H.2 Reproduction Workflow

I implemented the experiment as a Python pipeline covering data preparation, model inference, automated evaluation, manual-analysis support, statistical analysis, and figure generation. The main stages and their corresponding Python modules are listed in Table 10.

Table 10: Main experimental stages and corresponding Python modules.

Stage	Python module
Data preparation	<code>src.prepare_data</code>
Data inspection	<code>src.inspect_data</code>
Model inference	<code>src.run_inference</code>
Answer evaluation	<code>src.evaluate_answers</code>
Result aggregation	<code>src.combine_results</code>
Annotation sampling	<code>src.select_balanced_annotations</code>
Annotation analysis	<code>src.analyze_manual_annotations</code>
Statistical analysis	<code>src.statistical_analysis</code>
Qualitative examples	<code>src.select_qualitative_examples</code>
Figure generation	<code>src.create_paper_figures</code>

The workflow started with preparing and inspecting the benchmark. Model inference then generated the responses, which were processed for answer evaluation and combined for later analysis. After this, I selected the balanced sample used for manual annotation.

Once the manual review was complete, the annotation results were analyzed together with the statistical comparisons. The final steps selected the qualitative examples and generated the figures used in the paper.

Each module can be run using `python -m` followed by its module name. The project scripts and configuration files contain the required input and output paths, model settings, token limits, and other execution arguments. The hardware and generation settings used for the reported experiments are described in Appendix A.

H.3 Personal Research Contribution

I designed and carried out the experimental evaluation presented in this term paper. I selected the reasoning benchmarks and open-weight models, prepared the shared evaluation dataset, implemented the inference and evaluation pipeline, and collected the experimental results.

Because the available GPU had 4 GB of memory, I adapted the inference setup using 4-bit quantization. During preliminary testing, I also inspected incomplete DeepSeek responses and increased its maximum generation length from 256 to 768 new tokens so that both required output fields could be completed more reliably.

I evaluated answer correctness and strict-format compliance across all 900 generated responses. I then selected and manually reviewed the 90-response sample using the three explanation-quality dimensions and the primary failure categories described in Appendix C.

I also performed the statistical analysis, inspected the qualitative examples, and prepared the tables and figures used to present the results.

The main contribution of this work is an empirical comparison of answer performance and visible self-explanations under the same general experimental setup. The study combines automated answer evaluation with a structured manual analysis of explanation quality. It does not introduce a new language model, and the manual scores should not be interpreted as measurements of the causal faithfulness of the models’ internal reasoning.

Declaration of Academic Integrity

I,

Last name, first name:

Gavade Yash

Student ID number:

1757209

hereby declare that I have written this thesis independently and have not used any sources or aids other than those specified. All direct or indirect content and ideas from external sources are acknowledged as references. I confirm that I have not previously submitted this thesis in part or whole for any other academic credit nor has the work been submitted to any other examination office and has not been published.

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Programme:

M.Sc. Natural Language Processing

Examiner:

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Advanced Topics In Computational Text & Media Science

Place, Date

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